LED Matrix

LED1 OSO5PAS1C1A-1MA Orange 1mA
LED2
LED3
LED4
LED5
LED6
LED7
LED8
LED9
LED10
LED11
LED12
LED13
LED14
LED15
LED16
LED17
LED18
LED19
LED20

Ill populate these by hand because the led I need isnt on lcsc. Documentation in documentation parameter.

PWR

U7
LP5024RSMR
LED19 LED15 LED11 LED7 LED18 LED14 LED10 LED6 LED3 LED2 LED5 LED9 LED13 LED17
OUT0 OUT1 OUT2 OUT3 OUT4 OUT5 OUT6 OUT7 OUT8 OUT9 OUT10 OUT11 OUT12 OUT13 OUT14 OUT15
GND VCAP IREF EN SCL SDA VCC ADDR1 ADDR0 OUT23 OUT22 OUT21 OUT20 OUT19 OUT18 OUT17 OUT16
GND
1uF C576449 C7
R15 C852746 37k4
LED_EN
LED_SDA
LED_SCL
C8 C576449 1uF
PWR
VDD_nRF
R14 C105872 5k1
R16 C105872 5k1
R17 C138033 1M
LED20 LED16 LED12 LED8 LED4 LED1

MCU

U2
E73-2G4M08S1C
V Devider Battery measuring
PWR R4 C138033 1M
ADC
R5 C368359 2M49
GND
P1.11 P1.10 P0.03 AI4 GND P1.13 AI0 AI5 AI7 AI6
ADC COL5 ROW4 ROW2
NF2 P1.06 NF1 P1.04 SWC P1.02 SWD P1.00 P0.24 P0.22 P0.13 P0.20 D+ P17 D- P15 VBS RST
STDBY CHRG SWBCLK SWDIO COL1 COL2 COL3 COL4 D+ PWR_BTN_PRESS D- ROW1 VBUS RESET
Debug Connector
H1 C22373927
C5 C703694 4.7uF
PWR_BTN
R1 C138033 1M
PWR_BTN_PRESS
R2 C368359 2M49
XL1 P0.26 XL2 P0.06 AI3 P0.08 P1.09 AI2 VCC P12 GND P0.07 VDH GND DCH
ROW3 LED_EN LED_SDA LED_SCL VDD_nRF PWR
X1 C99010
C1 C182477 12pF
C3 C182477 12pF
L1 C223154 10uH
Reset Button
RESET
SW1 C720477
C60490 R8 10k
C6 C85923 100nF

ON/OFF Circuit

PWR_BTN
R6 3M48
C2 C528974
C94366 D1
C60491 R3 100k
U3 C2870716
CLK D Q GND VCC PRE CLR DFF
BAT+
10k C60490 R7
N.C A GND Y VCC Shmitt Trigger
SN74LVC1G17DBVR U1
PMOS_gate
C4 C85923 100nF
BAT+
PMOS_gate C389353 Q1
PWR

Key Matrix

COL1 COL2 COL3 COL4 COL5
ROW1 ROW2 ROW3 ROW4

S1 Choc HEX D2 C6341459	S4 Choc HEX D5 C6341459	S8 Choc HEX D9 C6341459	S12 Choc HEX D13 C6341459	S16 Choc HEX D17 C6341459
S2 Choc HEX D3 C6341459	S5 Choc HEX D6 C6341459	S9 Choc HEX D10 C6341459	S13 Choc HEX D14 C6341459	S17 Choc HEX D18 C6341459
S3 Choc HEX D4 C6341459	S6 Choc HEX D7 C6341459	S10 Choc HEX D11 C6341459	S14 Choc HEX D15 C6341459	S18 Choc HEX D19 C6341459
	S7 Choc HEX D8 C6341459	S11 Choc HEX D12 C6341459	S15 Choc HEX D16 C6341459	S19 Choc HEX D20 C6341459

BAT+
S20 Choc HEX
PWR_BTN

Charging and battery

USBC1
C165948
USB-C
GND VBUS SBU2 CC1 DN2 DP1 DN1 DP2 SBU1 CC2 VBUS GND
B12 B9 B8 A5 B7 D- A6 D+ A7 D- B6 D+ A8 B5 A9 B1 A12
EH EH EH EH
R9 C105872 5k1
R10 C105872 5k1
C2760275 F1
VBUS
U5
C2999147
R11 C274876 680
EP NC NC PROG GND VCC CHRG STDBY BAT
CHRG STDBY
R12 C60490 10k
R13 C60490 10k
VDD_nRF
BAT+
U4 C173752
Q2 C2830320
NC VDD VSS CO VM DO G1 D1/D2 G2 S1 D1/D2 S2
DW01A_C42380662 U6

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